0

OPERATING-POINT BOARD · SCENARIO CARD

Commission the board

The situation

Pump at 1 480 rpm with the full Ø174 impeller. Supply tank level gives a static head of 10 m; clean strainer, discharge valve fully open (K = 0.020). Place the pump curve, the system curve and the POR/AOR bar, mark the BEP, and mark the operating point.

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P1` `S2` `W1` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Where does the pump run relative to the BEP? Is it inside the POR?

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`

A

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Strainer fouls by 40 %

The situation

Debris blocks 40 % of the suction strainer area. Nothing else changes. Loss through a screen goes with 1/(open area)², so the strainer coefficient rises by (1/0.6)² = 2.8×: K goes from 0.020 to 0.029.

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P1` `S3` `W1` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Swap the system piece. Predict the new Q and H. Does the pump stay inside the POR? What happens to the suction pressure?

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`

B

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Supply tank level falls 5 m

The situation

The supply tank is being drawn down and its level drops 5 m. The pump lifts from the tank to the same delivery point, so the static head rises from 10 m to 15 m. Friction unchanged (K = 0.020).

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P1` `S2` `W1` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Slide the same system piece up 5 m. Predict Q and H. Compare the flow loss with scenario A: which change hurt more?

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`

C

OPERATING-POINT BOARD · SCENARIO CARD

Operator throttles the discharge to 18 m³/h

The situation

Production asks for 18 m³/h instead of 24. The operator closes the discharge valve until the flowmeter reads 18 m³/h. Static head 10 m; the extra loss is all in the valve.

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P1` `S4` `W1` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Which system piece crosses the pump curve at 18 m³/h? Predict the head. How much head is being burned in the valve, and what is the shaft power?

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`

D

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VFD reduces speed from 1 480 to 1 200 rpm

The situation

Instead of throttling, the same flow reduction is attempted with the VFD. Speed goes to 1 200 rpm; the valve stays fully open (K = 0.020, static head 10 m).

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P3` `S2` `W2` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Place the 1 200 rpm curve and move the POR/AOR bar and the BEP with it. Predict Q and H. Compare the shaft power with scenario C at the same flow.

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`

E

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Bonus — second pump started in parallel

The situation

A second identical pump is started in parallel to "double the flow". Baseline system (static head 10 m, K = 0.020), both at 1 480 rpm.

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P8` `S2` `W7` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Place the parallel curve. Predict the total flow and the flow per pump. Where does each pump sit in its POR?

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`

F

OPERATING-POINT BOARD · SCENARIO CARD

Bonus — trim the impeller for 18 m³/h

The situation

The 18 m³/h duty is permanent. Instead of throttling (C) or a VFD (D), the impeller is trimmed. Baseline system, 1 480 rpm.

On the board

1. Keep the baseline pieces in place unless the card changes them.
2. Pieces for this scenario: `P7` `S2` `W6` `BEP dot` `OP ? marker`
3. Register each piece to the grid (Q = 0 end and the white Q 10 / 20 / 30 ticks). Slide the system piece to the static head.
4. Put the BEP dot on the curve and lay the POR/AOR bar on the Q axis.
5. Place the red **?** marker where the curves cross.

Predict, then write it down

Which trimmed curve passes closest to 18 m³/h on the baseline system? Predict Q and H and the shaft power. What is the trim limit?

Flow Q (m³/h)	Head H (m)	% of Q_BEP	Inside POR?	η (%)	Shaft power (kW)
_____	_____	_____	yes / no	_____	_____

Then verify on the digital board or the hydraulic loop bench and mark the true point with the blue ✓. Power: P = ρ g Q H / η with Q in m³/s.

`Pump: H = 32 − 0.2 Q − 0.01 Q² at 1 480 rpm · Q_BEP 24 m³/h · System: H = H_s + K Q² · baseline H_s 10 m, K 0.020 · POR 70–120 % · AOR 50–130 %`